

TABLE I | Two service capacities across ten CIM technologies

INT8 logical payloads · Fixed reference organizations · Finite paired engineering scenarios

Technology / K × N	Optimistic				Typical				Pessimistic			
	ρ [MB/s]	τ [MB/s]	RI* [-]	U^* [-]	ρ [MB/s]	τ [MB/s]	RI* [-]	U^* [-]	ρ [MB/s]	τ [MB/s]	RI* [-]	U^* [-]
SRAM ACIM 128×128	57	3,200	0.018	2.3	44	3,200	0.014	1.8	27	1,600	0.017	2.1
SRAM DCIM 128×16	440	3,200	0.14	2.2	390	3,200	0.12	1.9	200	1,600	0.12	2.0
2D NOR 128×128	23	0.080	290	38,000	20	0.080	260	33,000	20	0.0091	2,100	270,000
3D NAND 4608×240	7.9	0.58	14	3,300	6.7	0.58	12	2,800	5.8	0.28	21	5,000
RRAM 128×64	29	4.7	6.1	390	29	4.3	6.7	430	29	3.5	8.3	530
MRAM 256×32	180	74	2.4	76	180	63	2.8	88	160	56	2.9	92
PCM 256×128	3.6	7.7	0.47	60	2.8	7.0	0.40	51	1.7	5.5	0.30	38
HfO ₂ FeRAM 128×128	36	240	0.15	20	22	99	0.23	29	13	54	0.24	31
Gain-cell eDRAM 64×64	26	170	0.16	9.9	14	160	0.089	5.7	9.6	140	0.071	4.5
3D FeFET 128×128	43	61	0.70	90	36	54	0.67	86	26	44	0.60	76

$\rho = B_S/\Delta_S$, $\tau = B_R/T_R$, $RI^* = \rho/\tau$, $U^* = T_R/\Delta_S = NRI^*$ for the selected one-Byte input and weight payloads.

Scenarios are finite paired engineering conditions, not statistical confidence intervals, same-chip PVT guarantees or strict bounds for all implementations.

Both capacities use complete service boundaries. Gain-cell is volatile and includes refresh; internal planes and maintenance do not increase logical payload.

Model-limited or unchanged budgets remain explicit; coincident values retain their source coordinates and do not establish zero uncertainty.

Native resources, sizes and precision conditions differ. 3D FeFET denotes vertical AND FeFET; see METHOD for actual varied inputs and remaining conditions.

Typical streaming and resident capacity

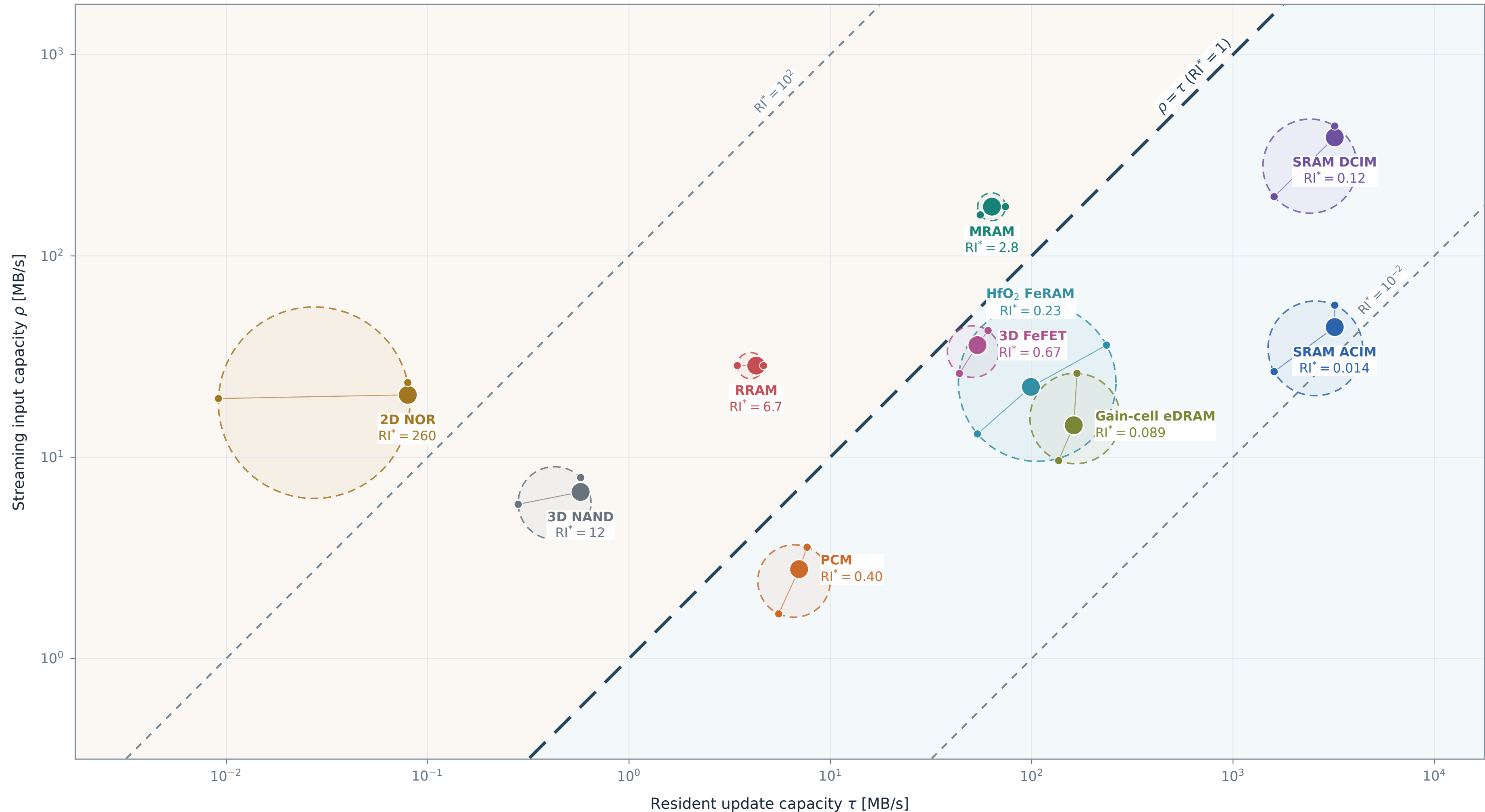
Selected reference INT8 payloads · $RI^* = \rho/\tau$



Native sizes and resources differ; these capacities do not rank equal-area or equal-work implementations.

Paired streaming and resident capacity

Large dots: typical · Small dots: optimistic / pessimistic · Paired scenarios · $RI^* = \rho/\tau$



Finite paired engineering scenarios; not confidence intervals, same-chip PVT guarantees or strict implementation bounds. Interiors may be infeasible.

Native sizes and resources differ; these capacities do not rank equal-area or equal-work implementations.